

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA1220	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Assessment Tool 1 – Application based questions
Weightage:	40%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
Student Name:	P. Reetha	Reg. No.:	192472396
Section / Batch:	2024	Date:	15/07/26

Instructions to Students

- Identify the relevant computer architecture concepts, functional units, or performance parameters required to solve the given application-based problem.
- Analyse the interaction between CPU, memory, bus structures, and I/O components in the given scenario.
- Apply appropriate instruction execution cycles, addressing modes, and architectural concepts to determine the correct solution.
- Compute performance metrics such as CPI, clock cycles, CPU execution time, speedup, or benchmarking measures using appropriate formulas and calculations.
- Interpret the calculated results and explain their impact on overall computer system performance.

QUESTIONS

1. A web developer notices that the displayed colour does not match the expected output because an incorrect hexadecimal colour code was entered. To identify the source of the mismatch, use an online colour picker or graphics software to compare two hexadecimal colour values (for example, #FF0000 and #F00000). Analyse how changing a hexadecimal value affects the binary representation and the displayed output with suitable illustration.
2. A programmer notices that a program produces different outputs when different input values are used. To examine the effect of varying input values, execute an 8085 program that performs arithmetic operations using different input values. Observe the changes in the Sign, Zero, Carry, and Parity flags after each operation. Analyse how the flag register influences program execution and determines why different inputs produce different flag values.
3. A user notices that a computer responds slowly while editing a document, browsing the internet, and playing audio simultaneously. To analyse the system behaviour, perform these activities together and observe the system's response. Analyse how the CPU, memory, and I/O devices cooperate to execute these tasks, and identify which functional unit is primarily responsible for processing, storing, and receiving data during the operations.

Code of Conduct [192472396]

I P. Reetha (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P. Reetha

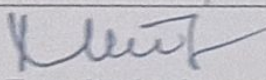
RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Q. No	Understanding of Problem Context (20)	Application of Concepts (30)	Problem-Solving Approach (20)	Accuracy of Solution (20)	Clarity (10)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 30	/ 20	/ 20	/ 10	/ 100

Course Coordinator


Faculty Incharge

CO1 – ASSESSMENT TOOL – 1

NAME : P. RISHITHA
REG NO: 192472396
COURSE NAME : COMPUTER ARCHITECTURE
COURSE CODE: CSA1220

1. ANALYSIS OF HEXADECIMAL COLOUR VALUES

AIM:

To compare two hexadecimal colour values and analyse how changing a hexadecimal value affects the binary representation and the displayed colour.

TOOLS USED:

- Online Colour Picker
- Binary Converter

COLOUR VALUES :

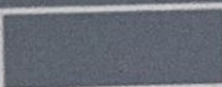
- Colour 1: #FF0000
- Colour 2: #F00000

PROCEDURE :

Open an online colour picker.

1. Enter the hexadecimal value #FF0000 and note its RGB and binary values.
2. Enter the hexadecimal value #F00000 and note its RGB and binary values.
3. Compare the binary representation and displayed colours.

OBSERVATION

Comparison of #FF0000 and #F00000		
Property	#FF0000	#F00000
Colour Preview		
RGB Value	rgb(255, 0, 0)	rgb(240, 0, 0)
Binary (R)	11111111	11110000
Binary (G)	00000000	00000000
Binary (B)	00000000	00000000
Description	Pure Bright Red	Slightly Darker Red

SHORT EXPLANATION

Changing the hexadecimal value from FF to F0 changes the binary value of the red component from 11111111 to 11110000. This reduces the red intensity from 255 to 240, producing a slightly darker shade of red.

FIGURE 1 : COLOUR PICKER OUTPUT

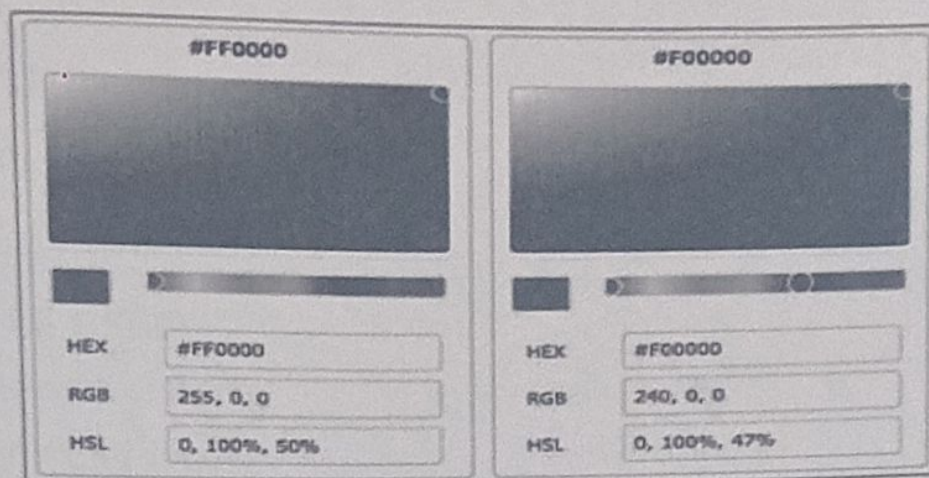


FIGURE 2 : BINARY REPRESENTATION

Hex Digit	F	F	0	0	0	0
Binary	1111	1111	0000	0000	0000	0000
Hex Digit	F	0	0	0	0	0
Binary	1111	0000	0000	0000	0000	0000

RESULT

Small changes in hexadecimal values directly affect the binary representation, RGB values, and displayed colour.

2. EFFECT OF INPUT VALUES ON 8085 FLAGS

AIM:

To execute an 8085 arithmetic program using different input values and observe the Sign, Zero, Carry and Parity flags.

PROGRAM

MVI A,25H

MVI B,15H

ADD B

HLT

TOOLS USED

8085 Simulator

OBSERVATION

Test	Input A	Input B	Result	S	Z	P	CY
1	25H	15H	3AH	0	0	1	0
2	80H	80H	00H	0	1	1	1
3	7FH	01H	80H	1	0	0	0
4	FFH	01H	00H	0	1	1	1

SHORT EXPLANATION

Different input values produce different arithmetic results. The flag register is updated after every operation and indicates whether the result is negative, zero, has a carry, or has even parity. These flags influence conditional branching and program execution.

TEST CASES:

Test Case 1: 25H + 15H = 3AH					
Registers		Flag Register (F)			
A	3A	S	Z	AC	P
B	15				
C	00	0	0	0	1
D	00				
E	00				
H	00				
L	00				
		CY			

Test Case 2: 80H + 80H = 00H					
Registers		Flag Register (F)			
A	3A	S	Z	AC	P
B	80				
C	00	0	1	1	1
D	00				
E	00				
H	00				
L	00				
		CY			

Test Case 3: 7FH + 01H = 80H					
Registers		Flag Register (F)			
A	80	S	Z	AC	P
B	01				
C	00	1	0	1	0
D	00				
E	00				
H	00				
L	00				
		CY			

Test Case 4: FFH + 01H = 00H					
Registers		Flag Register (F)			
A	00	S	Z	AC	P
B	01				
C	00	0	1	1	1
D	00				
E	00				
H	00				
L	00				
		CY			

RESULT

The Sign, Zero, Carry and Parity flags change according to the arithmetic result, enabling the processor to make decisions during program execution.

3. ANALYSIS OF SYSTEM BEHAVIOUR DURING MULTIPLE TASKS

AIM

To perform multiple activities simultaneously and analyse how the CPU, memory and I/O devices cooperate during execution.

ACTIVITIES PERFORMED

- ▶ Editing a document in LibreOffice Writer
- ▶ Browsing the Internet
- ▶ Playing audio

OBSERVATION

Functional Unit Responsibility

CPU	Executes instructions and processes data
Memory (RAM)	Stores programs and temporary data
I/O Devices	Receive input and provide output

SHORT EXPLANATION

When multiple applications run simultaneously, the CPU rapidly switches between tasks, RAM stores the required data for each application, and I/O devices handle keyboard, mouse, display, speakers and network communication. Together, these functional units ensure smooth multitasking.

FIGURE 1: MULTIPLE APPLICATIONS RUNNING

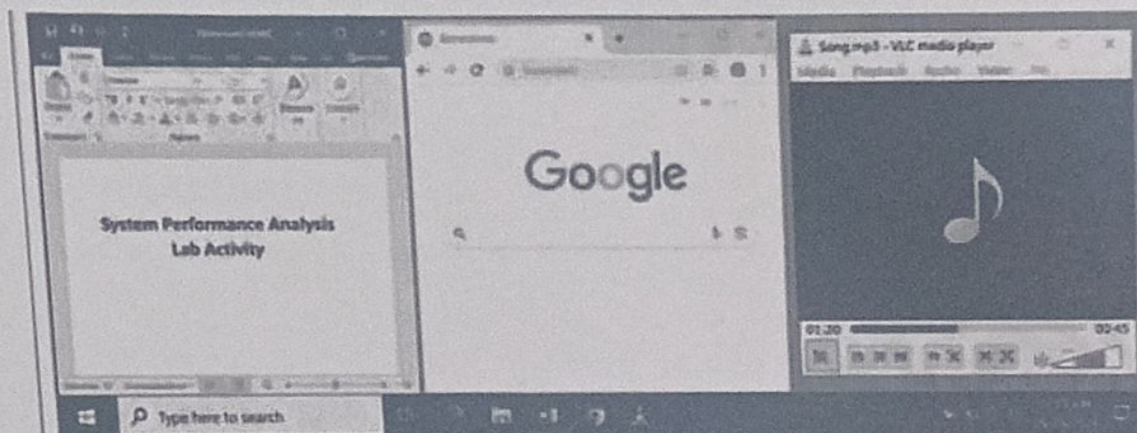


FIGURE 2: TASK MANAGER (PERFORMANCE TASK)

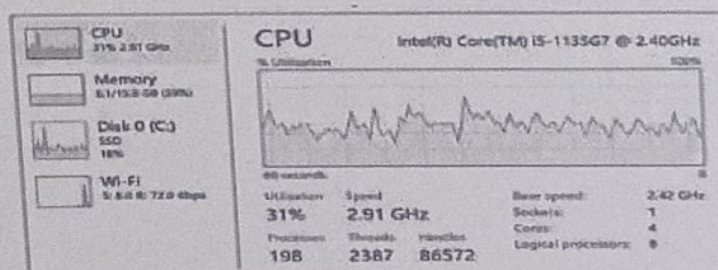


FIGURE 3 : SYSTEM RESOURCE USAGE (VALUES MAY VARY)

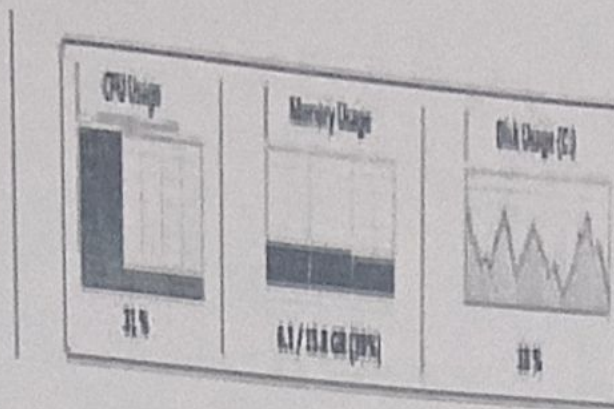


FIGURE 4 : FUNCTIONAL UNITS INVOLVED

